Loan Prediction Using Machine Learning

Data Science Internship Report

Name: Prathamesh Borkar

Task: Loan Prediction Using Machine Learning

Executive Summary

This project aims to build a machine learning model capable of predicting whether a loan application will be approved based on applicant information such as income, education, marital status, loan amount, credit history, and property area.

The dataset contains **614 loan applications** with **11 input features** and **1 target variable (Loan_Status)**. Missing values were handled using median and mode imputation. Categorical variables were converted into numerical values using Label Encoding. Exploratory Data Analysis (EDA) was performed using histograms, boxplots, count plots, and a correlation heatmap to understand the data distribution and relationships among variables.

The dataset was divided into training and testing sets using an 80:20 split. Numerical features were standardized using StandardScaler. Since the dataset was slightly imbalanced, class weights were applied during model training.

Two classification algorithms were trained and evaluated:

- Logistic Regression
- Random Forest Classifier

Random Forest achieved the highest overall accuracy and F1-score, making it the preferred model for this problem.

Introduction

Objective

The objective of this project is to build a machine learning model capable of predicting whether a loan application will be approved or rejected using applicant details such as income, education, marital status, credit history, and loan amount.

Dataset Description

- Dataset Name: Loan Prediction Dataset
- Total Records: 614
- Features: 11
- Target Variable: Loan_Status

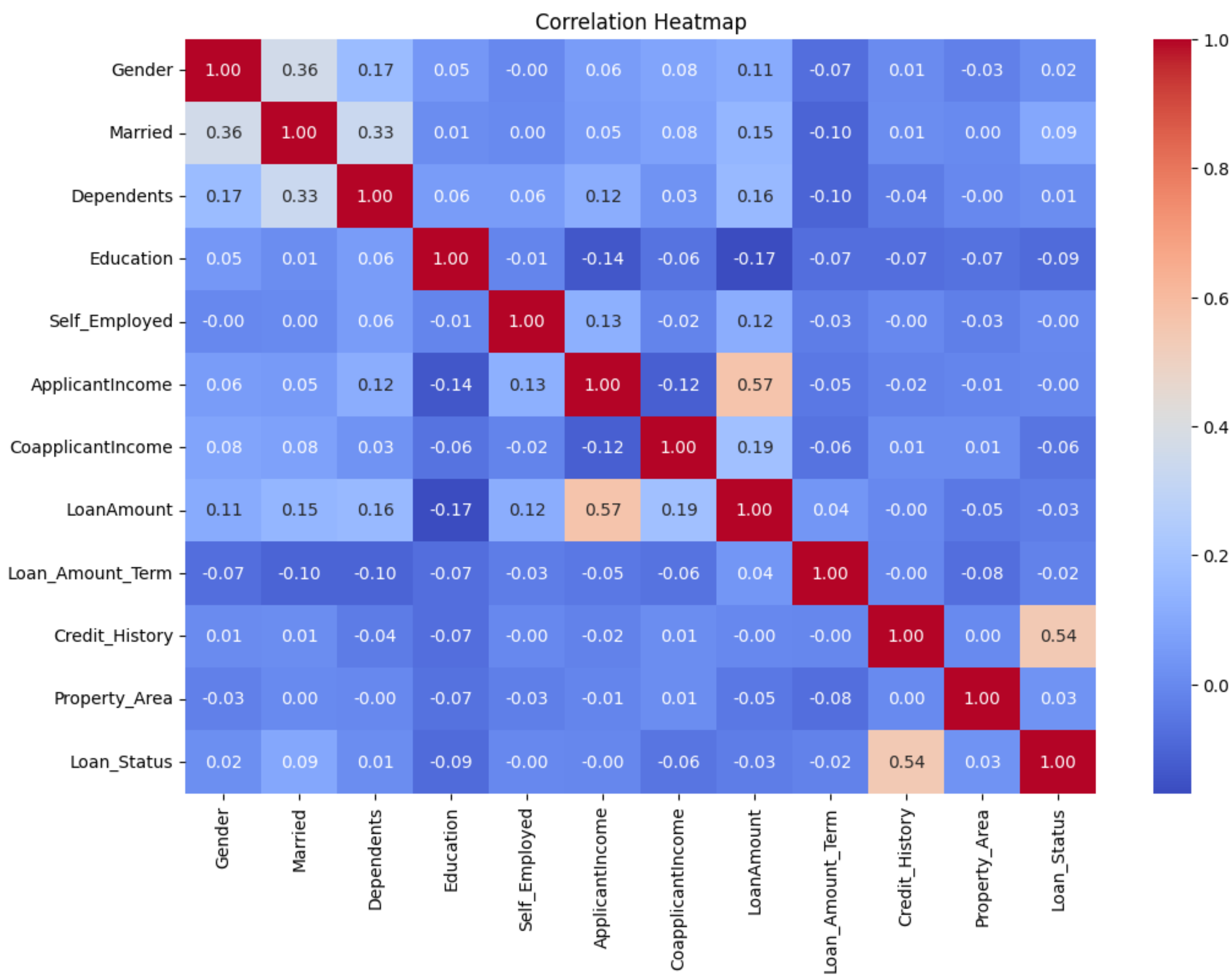
Exploratory Data Analysis (EDA)

Loan Approval Distribution

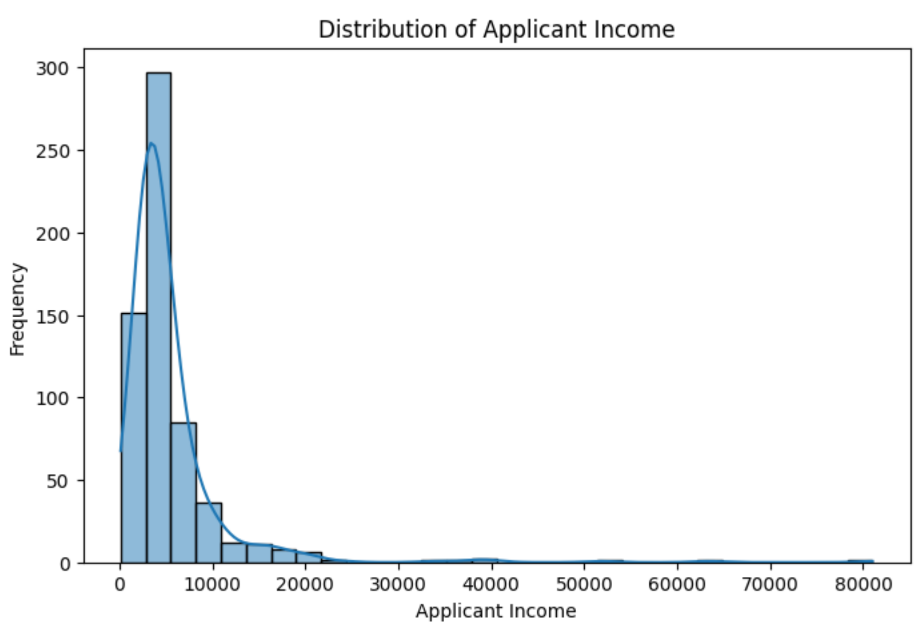
The dataset is slightly imbalanced. Approximately 69% of loan applications were approved, while 31% were rejected.

Correlation Heatmap

The correlation heatmap shows that Credit History has the strongest positive relationship with Loan Approval, making it one of the most important features

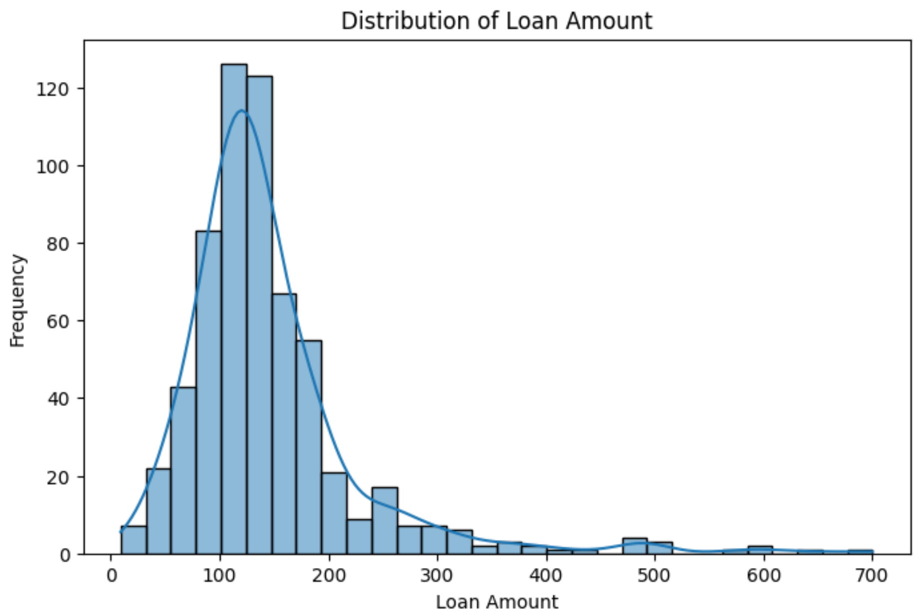


Applicant Income Distribution



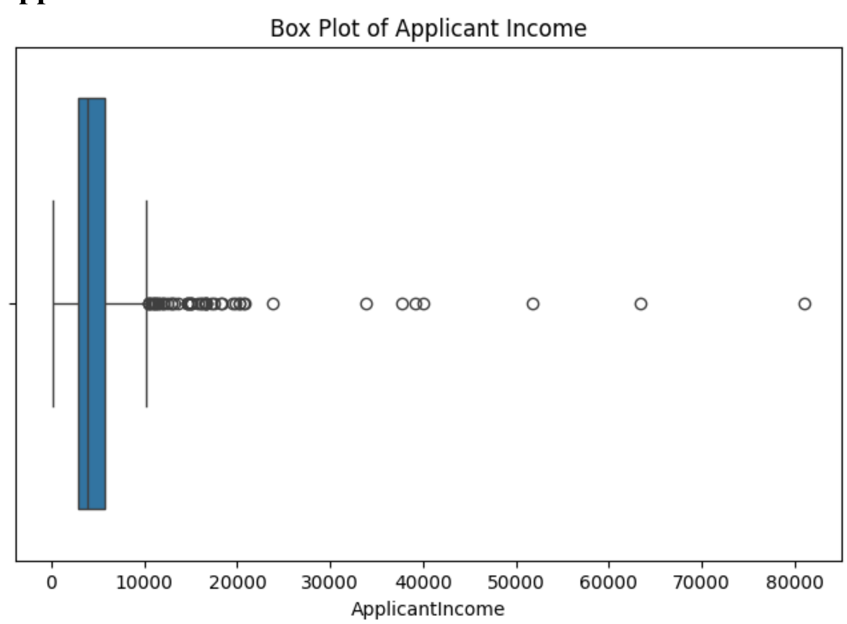
The Applicant Income distribution is highly right-skewed with several high-income outliers.

Loan Amount Distribution



Most loan amounts are concentrated between 100 and 170 units, with a few large loan outliers.

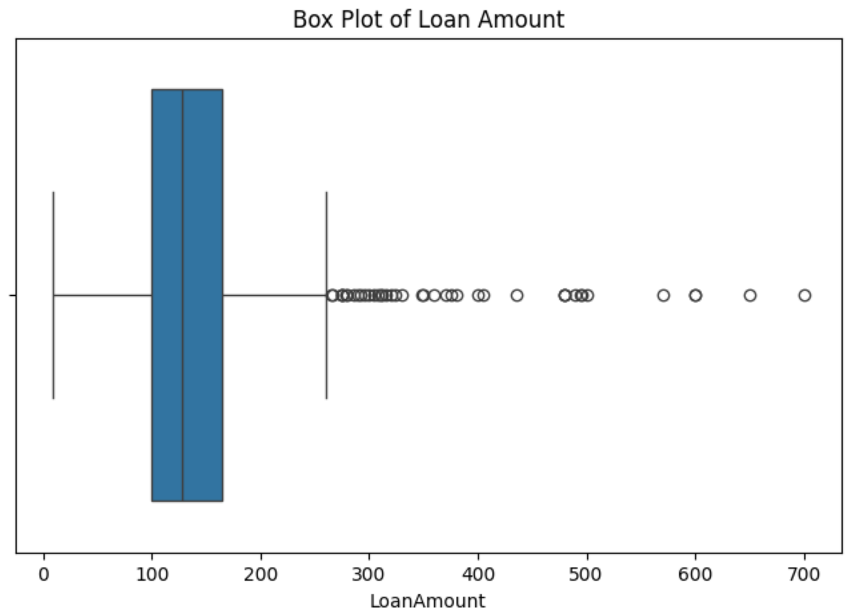
Applicant Income Box Plot



The box plot reveals numerous outliers in applicant income, indicating a few applicants earn significantly higher incomes than the majority.

Loan Amount Box Plot

Loan Amount also contains several outliers, although most applications fall within a moderate range.



Data Preprocessing

The following preprocessing steps were performed:

- Removed Loan_ID
- Filled missing values
- Label Encoding
- Feature Scaling
- Train-Test Split
- Class Weight Balancing

Machine Learning Models

Logistic Regression

Logistic Regression is a linear classification algorithm used as the baseline model.

Performance

Accuracy: 80.49%

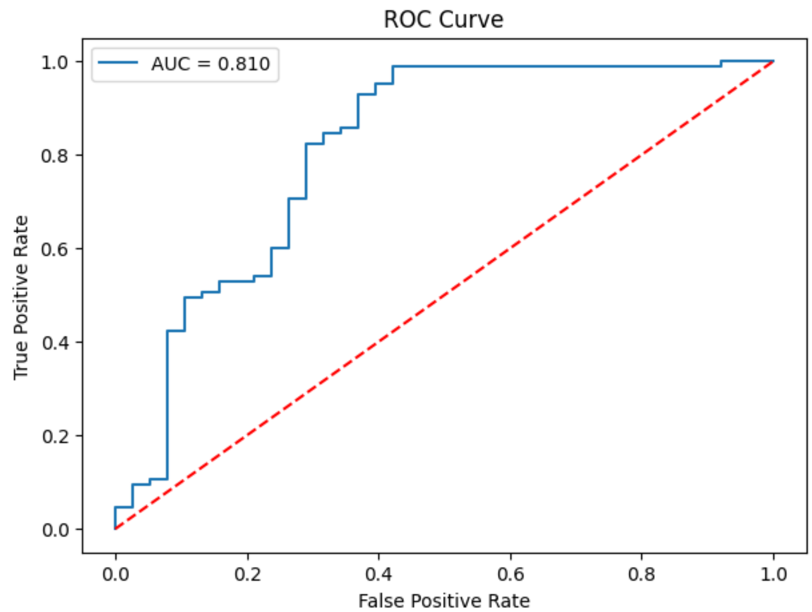
Precision: 84.27%

Recall: 88.24%

F1 Score: 86.21%

ROC-AUC: 80.96%

ROC Curve



The ROC curve achieved an AUC score of 0.81, indicating good discrimination between approved and rejected loans

Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Performance

Accuracy: 83.74%

Precision: 84%

Recall: 94%

F1 Score: 89%

ROC-AUC: 79.60%

Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	80.49 %	83.74 %
Precision	84.27 %	84.00 %
Recall	88.24 %	94.00 %
F1 Score	86.21 %	89.00 %
ROC-AUC	80.96 %	79.60 %

Model Trade-offs

Logistic Regression

- Easy to interpret
- Fast training
- Good baseline model
- Slightly better ROC-AUC

Random Forest

- Highest Accuracy
- Highest Recall
- Better F1 Score
- Better overall prediction performance
- Less interpretable than Logistic Regression

Suggested Threshold for Deployment

The default probability threshold of **0.50** is recommended for deployment because it provides a good balance between precision and recall.

For banking systems that prioritize reducing risky loan approvals, the threshold can be increased to **0.60**

Conclusion

Random Forest outperformed Logistic Regression by achieving an accuracy of **83.74%** and an F1-score of **89%**. Therefore, Random Forest is selected as the final model for loan approval prediction.